

1984, three years after the release of the IBM PC (the first mass-market computer). The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption thus far suggests that it has the potential to be similarly disruptive. Moreover, we show that genAI adoption by workers is proceeding at a faster pace than adoption by firms, highlighting both challenges and opportunities for managers seeking to capitalize on this new technology (Babina et al., 2024; Bonney et al., 2024; Doshi and Moore, 2025; McElheran et al., 2024).²

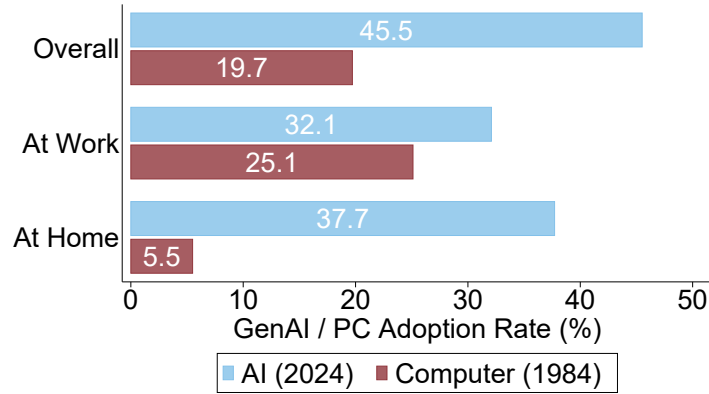
Second, which workers are currently using genAI, and what factors contribute to an individual’s adoption decision? We find substantial variation in genAI adoption across demographic and labor market characteristics. Adoption has been faster for male, younger, more educated, and higher-wage workers. Adoption also differs greatly by occupation, suggesting that differences in job roles and responsibilities are an important determinant of adoption. Consistent with this, we show that recent estimates of genAI task-based exposure are highly predictive of actual adoption (Eloundou et al., 2024; Felten et al., 2021), and that occupations with lower adoption also have higher shares of workers who report that genAI is not useful for their job. These findings contribute to our understanding of how genAI maps onto specific tasks and roles within the firm, which is relevant both for allocating adoption support and adapting firm organization.

We provide evidence that firm climate and policies also appear to play an important role in workers’ adoption choice. A sizable share of workers who do not use genAI cite workplace barriers as a key reason, and this share is particularly large in a few occupations who may deal with sensitive data. Alternatively, workers whose employers encourage them to use genAI are much more likely to do so, even after controlling for other characteristics. This suggests that when job tasks are amenable to genAI adoption, firms can play an active role by promoting usage, offering training, and removing organizational barriers to adoption.

Third, how large are potential productivity gains from genAI? Based on self-reported time savings due to genAI, the average genAI user would need to work 5.2% more hours in order to complete their work without genAI, implying 1.4% time savings across all workers (including non-users). We use a standard production model to estimate potential productivity gains at the industry and aggregate level. We find wide variation in potential productivity gains by industry,

²The Census BTOS Survey estimates between August and November 2024 5.4% of firms had incorporated AI technologies into their production process. This suggests that firm-based measures may miss an important share of genAI adoption. Aaron Levie, the founder and CEO of Box, a cloud storage provider, compares this pattern with past technologies: “It’s almost a one-to-one parallel with the early wave of consumerization of IT. We saw that exact same dynamic in the early 2010s, which was employees of enterprises were pretty frustrated with their traditional document management and collaboration technology, and so they would go out and just go onto a web browser, sign up for a Box account, and then bring it into their organization.” (Levie, 2024)

Figure 2: Computer and Generative AI Adoption 2-3 Years After First Mass-Market Product



Notes: The figure displays usage rates for computers (PCs) and genAI in 1984 and 2024, respectively. The IBM PC was released in 1981, which we refer to as the first mass-market computer, implying that 1984 is three years after the first mass-market computer. ChatGPT was released in November 2022, which we refer to as the first mass-market genAI product, implying the fall of 2024 is two years after the first mass-market product for that technology. The data source for genAI is the August and November 2024 wave of the RPS (blue bars). The data source for computers is the 1984 Computer and Internet Use Supplement of the CPS (red bars). The sample for the Overall and At Home bars in the RPS and CPS is all individuals ages 18-64 (RPS, $N = 9726$; CPS, $N = 61,708$). The sample for the For Work bars is employed individuals ages 18-64 (RPS, $N = 6935$; CPS, $N = 89,271$).

measured using very similar questions for both technologies.) The first mass-market computer was the IBM PC, which was released in August 1981. The first mass-market genAI product was ChatGPT, which was released in November 2022.⁸ For the PC, we measure adoption in the CPS CIU for the first time in 1984, i.e. three years after the first mass-market computer. The RPS provides adoption estimates for 2024, two years after the first mass-market genAI product.

The top pair of bars in Figure 2 refers to overall adoption for the two technologies two to three years after the introduction of the first mass-market product. The bar for the RPS repeats the 45% overall adoption rate for genAI reported in Figure 1a. Overall PC adoption is roughly 20% in 1984, implying that genAI has been adopted at a faster pace than the PC. As another more recent comparison, in 1995 the internet was opened to commercial traffic after the decommissioning of NSFNet (Leiner et al., 2009).⁹ Two years later, 21% of US adults reported access to the internet according to the International Telecommunication Union, which compiles subscriber data from national regulatory bodies and service providers (Pena-Lopez et al., 2009). This is a similar pace of overall adoption to PCs, and substantially slower than the adoption of genAI.

⁸Three notable computer products had already been released in 1977 (the Apple II, the Commodore PET, and the TRS-80). Choosing earlier introduction dates would imply slower adoption rates for a given technology. The 1981 IBM PC was the first computer to sell over one million units (Abbate, 1999). We chose these dates based in part on ChatGPT’s answers to questions about the year of the first mass market product for each technology. The exact prompts and answers are available upon request.

⁹1995 was also the year of Netscape’s initial public offering and the year that AOL 3.0 was released.

rapidly (Nelson and Phelps, 1966). Third, genAI may directly increase workers’ productivity and wages. Our data do not allow us to distinguish between these explanations.

4.1 Tasks, Predicted Exposure, and Generative AI Adoption

Why do some groups of workers adopt genAI at higher rates than others? One reason could be that workers differ in their job tasks or responsibilities, and genAI is more helpful in assisting some tasks than others.

Consistent with this explanation, we first document that genAI adoption differs greatly between occupations; see Figure 5. The three occupations with the highest genAI adoption are computer and mathematical, management, and business and finance occupations (54%, 52%, and 48%, respectively) and the bottom three are blue collar, office and administration, and personal services occupations (26%, 23%, and 16%, respectively). At the same time, genAI adoption exceeds 20% in all but one occupation groups, suggesting that the technology is useful for a broad range of occupations. As with demographic differences, differences between occupations are driven by the rates of daily and weekly genAI users.

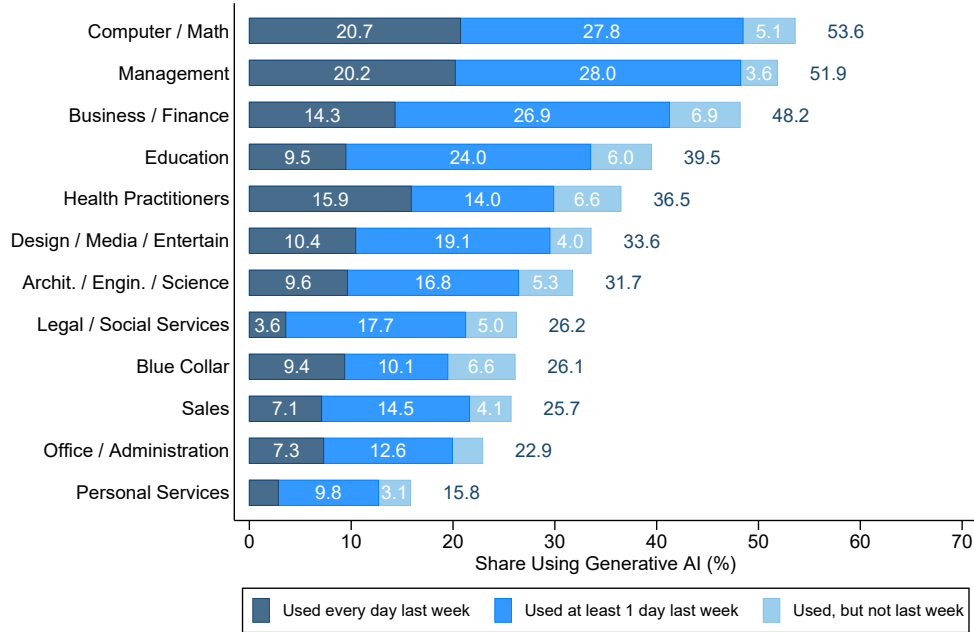
Do different adoption rates between occupations reflect differences in tasks or responsibilities? One approach to assess this is to construct task-based-exposure measures across occupations (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020). For example, Eloundou et al. (2024) constructs a rubric that assigns each task in the O*NET database a zero or one corresponding to whether LLMs (a popular class of genAI) could feasibly produce large productivity gains for that task. They then construct a predicted exposure score for each occupation by aggregating task-level scores for each occupation. Using RPS data, we can ask whether these predictions are correlated with adoption. Our study is the first to evaluate these task-based “exposure” predictions using actual genAI adoption rates.¹²

Figure 6a plots an occupation group’s predicted genAI exposure from Eloundou et al. (2024) against actual genAI adoption in the 2024 RPS. We find that predicted genAI exposure is highly correlated with actual genAI adoption ($\rho = 0.65$). In Figure 6b, we find a similar correlation ($\rho = 0.71$) using the predicted exposure score from Felten et al. (2021). For example, personal service and blue-collar occupations have relatively low predicted exposure and also relatively low adoption, while computer and mathematical occupations have both high predicted exposure and high adoption.

The high correlation between predicted exposure and actual adoption provides partial validation of task-based exposure predictions, supporting their utility for researchers. At the same

¹²Handa et al. (2025) assign tasks to conversations by Claude.ai Free and Pro users and show that task usage rates are correlated with the predictions by Eloundou et al. (2024).

Figure 5: Generative AI Work Adoption By Occupation



Notes: The figure shows adoption rates at work for genAI by occupation. The data source is the August and November 2024 waves of the RPS (blue bars). The sample is employed individuals age 18-64 (RPS, $N = 6883$). We exclude workers in “agriculture / forestry / fishing” due to a very low sample size. For details on occupation categories, see Appendix A.4.

time, we also note particular occupations whose adoption rates deviate from what we would expect based on their exposure score. In particular, office and administrative occupations and managers are predicted to be similarly exposed to genAI, but actual adoption by managers is more than double that of office and administrative workers. One potential reason for these discrepancies could be that existing task-based predictions do not perfectly identify which occupations would benefit from genAI.

To investigate this possibility, we leverage additional information about workers’ adoption decisions from the RPS. Among users who did not adopt genAI, we provided a list of potential reasons why they did not adopt. We then asked respondents to select the reasons that were relevant for their decision and to rank them in order of importance. We define the share of an occupation who does not use genAI because it is not useful for their job as the share of all workers in that occupation who ranked “GenAI cannot help me with my job” in their top two reasons for not using genAI. (For the full list of potential reasons, see Appendix A.5.)

Figure 6c plots genAI adoption against the share who reported that it was not useful. We find that occupational adoption is highly correlated (-0.88) with the share who reported genAI was not useful. For example, only 7% of workers in computer / math occupations reported that genAI was not useful for their job, compared with 30% of personal service workers.